

Table 1: Bug detection results in 24 hours. For each test case, $\hat{A}_{12}$ values with statistical significance ($p < 0.05$) are **bolded**; T.O. indicates the bugs is not triggered within the time budget. The best results in each line are highlighted in **green**.

Program	Bug ID	PartitionFuzz			AFL++			AFLEdge			glibFuzzer			UltraFuzz			AFLTeam			PartitionFuzz-P			PartitionFuzz-E			PartitionFuzz-S		
		TTE	Runs	$\hat{A}_{12}$	TTE	Runs	$\hat{A}_{12}$	TTE	Runs	$\hat{A}_{12}$	TTE	Runs	$\hat{A}_{12}$	TTE	Runs	$\hat{A}_{12}$	TTE	Runs	$\hat{A}_{12}$	TTE	Runs	$\hat{A}_{12}$	TTE	Runs	$\hat{A}_{12}$	TTE	Runs	$\hat{A}_{12}$
sqlite3	SQL002	21h55m	1		T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	24h0m	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55
	SQL006	23h33m	1		T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	24h0m	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55
	SQL007	22h4m	2		T.O.	0	0.60	23h24m	1	0.56	T.O.	0	0.60	T.O.	0	0.60	24h0m	0	0.60	23h5m	1	0.56	T.O.	0	0.60	T.O.	0	0.60
	SQL013	11h44m	7		18h31m	5	0.78	14h56m	7	0.72	T.O.	0	0.85	17h29m	4	0.76	20h28m	3	0.81	12h58m	5	0.57	16h32m	5	0.65	18h6m	4	0.79
	SQL018	9h8m	9		9h47m	8	0.76	17h30m	5	0.93	20h17m	2	0.94	23h40m	2	0.94	20h23m	2	0.94	11h5m	7	0.79	15h26m	5	0.88	13h4m	6	0.83
	SQL020	22h9m	1		T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	24h0m	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55
poppler	PDF002	21h46m	1		T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	24h0m	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55
	PDF004	T.O.	0		T.O.	0	0.50	T.O.	0	0.50	T.O.	0	0.50	T.O.	0	0.50	24h0m	0	0.50	23h51m	1	0.45	T.O.	0	0.50	T.O.	0	0.50
	PDF008	21h43m	1		T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	24h0m	0	0.55	21h49m	1	0.51	T.O.	0	0.55	22h15m	1	0.51
	PDF010	17m38s	10		41m36s	10	0.97	1h34m	10	1.00	1h54m	10	1.00	2h53m	10	1.00	5h44m	10	1.00	41m54s	10	0.89	1h24m	10	1.00	5h32m	8	0.93
	PDF016	17h53m	5		20h52m	4	0.65	23h20m	1	0.73	22h0m	1	0.73	T.O.	0	0.75	23h47m	1	0.73	19h34m	4	0.65	20h51m	4	0.54	22h21m	1	0.73
	PDF018	8h32m	10		12h28m	8	0.95	9h45m	10	0.67	20h5m	2	0.87	10h54m	7	0.72	10h30m	9	0.76	12h28m	6	0.77	12h59m	10	0.79	12h43m	7	0.76
	PDF019	22h52m	1		T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	24h0m	0	0.55	21h50m	1	0.50	T.O.	0	0.55	22h57m	1	0.51
	PDF021	20h33m	3		T.O.	0	0.65	T.O.	0	0.65	T.O.	0	0.65	T.O.	0	0.65	24h0m	0	0.50	T.O.	0	0.65	21h48m	1	0.60	T.O.	0	0.65
	LUA004	12h35m	7		17h39m	6	0.76	18h14m	4	0.79	T.O.	0	0.85	18h24m	4	0.75	17h37m	4	0.70	14h48m	5	0.61	18h3m	3	0.76	17h7m	5	0.69
libxml2	XML009	22h2m	1		T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55	N/A	N/A	N/A	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55
	XML017	18h37m	7		19h53m	6	0.76	T.O.	0	0.85	T.O.	0	0.85	T.O.	0	0.85	N/A	N/A	N/A	21h52m	4	0.77	22h35m	4	0.73	T.O.	0	0.85
libtiff	TIF001	20h7m	3		21h58m	2	0.58	19h52m	2	0.54	T.O.	0	0.65	T.O.	0	0.65	23h52m	1	0.62	21h13m	4	0.56	T.O.	0	0.65	T.O.	0	0.65
	TIF002	4h16m	10		9h22m	8	1.00	6h41m	10	0.94	T.O.	0	1.00	22h5m	2	1.00	16h39m	6	1.00	6h20m	10	0.74	6h30m	10	0.69	5h53m	10	0.65
	TIF007	17h41m	4		T.O.	0	0.70	23h57m	1	0.67	1h44m	10	0.03	T.O.	0	0.70	20h56m	2	0.64	22h40m	1	0.66	17h29m	3	0.51	18h21m	6	0.48
	TIF008	6h1m	8		13h53m	7	0.83	13h33m	7	0.83	T.O.	0	0.90	23h49m	1	0.89	15h13m	5	0.85	10h55m	8	0.63	11h14m	6	0.69	10h31m	7	0.65
	TIF012	21h57m	1		T.O.	0	0.55	23h14m	1	0.51	T.O.	0	0.55	23h51m	1	0.51	24h0m	0	0.55	T.O.	0	0.55	T.O.	0	0.55	T.O.	0	0.55
	TIF014	T.O.	0		T.O.	0	0.50	T.O.	0	0.50	2h29m	10	0.00	T.O.	0	0.50	24h0m	0	0.50	T.O.	0	0.50	T.O.	0	0.50	T.O.	0	0.50
	libsndfile	1m4s	10		1m13s	10	0.53	1m6s	10	0.51	9m59s	10	0.95	2m20s	10	0.67	2m37s	10	0.62	3m51s	10	0.76	3m37s	10	0.71	5m12s	10	0.82
	libpng	11m48s	10		13m50s	10	0.30	23m10s	10	0.79	13m25s	10	0.52	15m27s	10	0.62	32m8s	10	0.93	17m15s	10	0.68	17m37s	10	0.66	13m43s	10	0.58
	libpng	1m31s	10		39s	10	0.31	46s	10	0.44	4m40s	10	0.89	1m16s	10	0.52	44s	10	0.48	2m32s	10	0.66	3m6s	10	0.71	3m33s	10	0.81
libpng	PNG001	9h26m	9		11h54m	8	0.68	12h31m	6	0.75	22h16m	1	0.95	19h20m	4	0.87	20h45m	2	0.91	9h52m	9	0.49	12h58m	7	0.61	15h35m	7	0.70
	PNG003	T.O.	0		T.O.	0	0.50	T.O.	0	0.50	T.O.	0	0.50	24h0m	0	0.50	24h0m	0	0.50	T.O.	0	0.50	T.O.	0	0.50	T.O.	0	0.50
	PNG006	T.O.	0		T.O.	0	0.50	T.O.	0	0.50	4h36m	10	0.00	T.O.	0	0.50	24h0m	0	0.50	T.O.	0	0.50	T.O.	0	0.50	T.O.	0	0.50
	PNG007	11m34s	10		11m34s	10	0.55	10m1s	10	0.51	2h41m	10	0.70	1h45m	10	0.73	23m47s	10	0.60	12m4s	10	0.65	13m19s	10	0.67	11m52s	10	0.58
Mean		15h6m	4.73		17h14m	3.73	0.63	17h22m	3.50	0.64	16h55m	3.20	0.63	19h4m	2.50	0.67	18h10m	3.04	0.66	16h31m	3.83	0.61	17h8m	3.50	0.63	17h21m	3.43	0.63
Total Bugs (30)		26			15 (1.73x)			17 (1.53x)			13 (2x)			13 (2x)			15 (1.73x)			20 (1.3x)			16 (1.63x)			16 (1.63x)		

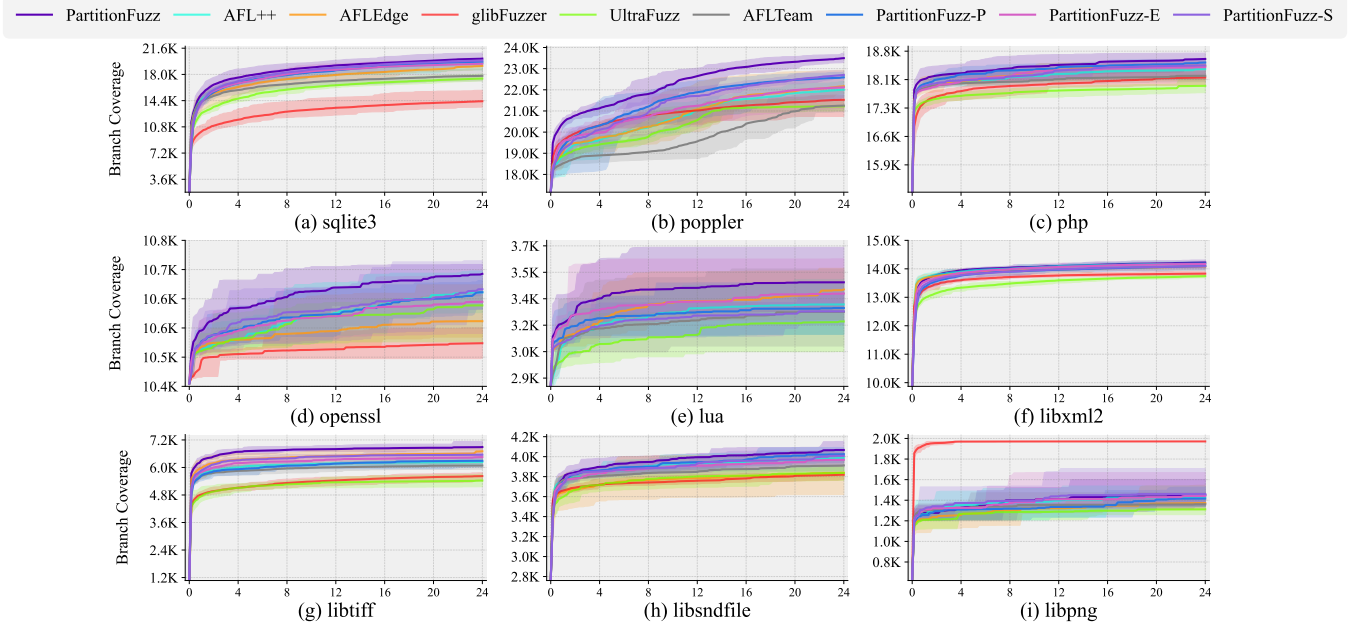


Figure 1: Branch coverage found by evaluated fuzzers with 4 cores in 24h.

Table 2: Statistics of branch coverage. For each test case, the best result and $\hat{A}_{12}$ values with statistical significance ($p < 0.05$) are **bolded**.

Program	PartitionFuzz		AFL++		AFLEdge		glibFuzzer		UltraFuzz		AFLTeam		PartitionFuzz-P		PartitionFuzz-E		PartitionFuzz-S	
	#BrCov	$\hat{A}_{12}$	#BrCov	$\hat{A}_{12}$	#BrCov	$\hat{A}_{12}$	#BrCov	$\hat{A}_{12}$	#BrCov	$\hat{A}_{12}$	#BrCov	$\hat{A}_{12}$	#BrCov	$\hat{A}_{12}$	#BrCov	$\hat{A}_{12}$	#BrCov	$\hat{A}_{12}$
sqlite3	20169	0.82	19557	0.97	19174	1.00	14313	1.00	17434	1.00	17801	1.00	19745	0.76	19470	0.94	19914	0.60
poppler	23496	1.00	22005	1.00	22171	1.00	21525	1.00	21216	1.00	21252	1.00	22588	1.00	22118	1.00	22710	1.00
php	18574	0.92	18316	0.89	18404	0.89	18104	1.00	17891	1.00	18147	0.99	18484	0.80	18367	0.92	18409	0.88
openssl	10708	0.87	10657	0.99	10579	0.99	10518	1.00	10622	1.00	N/A	N/A	10658	0.90	10632	0.96	10666	0.79
lua	3459	0.76	3326	0.58	3416	0.58	N/A	N/A	3223	0.97	3281	0.82	3304	0.84	3392	0.64	3284	0.93
libxml2	14221	0.67	14184	0.67	14081	0.99	13832	1.00	13741	1.00	N/A	N/A	14096	0.93	14170	0.71	14094	0.94
libtiff	6890	1.00	6237	0.71	6706	0.71	5623	1.00	5430	1.00	6084	1.00	6290	1.00	6450	1.00	6554	1.00
libsndfile	4067	0.83	3988	0.95	3839	0.95	3819	1.00	3838	1.00	3912	0.99	4025	0.73	3968	0.93	4009	0.83
libpng	1451	0.55	1429	0.55	1373	0.78	1971	0.00	1311	0.96	1361	0.84	1416	0.58	1441	0.55	1460	0.55